



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MD Razia
Student Name	Ritwik
Roll Number	2520080051
Branch	AI&DS
Course Outcome	CO - 1

WaterGrid – Smart Water Distribution & Monitoring System

Work Done Summary:

In this case study, we implemented WaterGrid, a Smart Water Distribution & Monitoring System using a self-balancing AVL Tree to manage pipeline pressure nodes. The AVL Tree stores pipeline IDs in sorted order and automatically maintains balance after every insertion. We inserted multiple pipeline records and observed balancing operations such as LL, RR, LR, and RL rotations. We also implemented a node-search operation to retrieve pipeline pressure information efficiently.

The AVL Tree stores movie titles in sorted order and automatically maintains balance after every insertion.

We inserted multiple pipeline records into the WaterGrid AVL Tree and observed balancing operations such as LL, RR, LR, and RL rotations. We also implemented a pipeline search operation to retrieve water pressure data efficiently.

Implementation Details:

1. Created an AVL Tree structure for storing WaterGrid pipeline pressure nodes.
2. Inserted pipeline records into the AVL Tree for balanced distribution management.
3. Calculated balance factors after every pipeline insertion to maintain tree height.
4. Performed AVL rotations whenever the pressure-node tree became unbalanced.
5. Maintained a balanced tree structure to ensure $O(\log n)$ access for all nodes.
6. Implemented search functionality to locate pipeline pressure data quickly.
7. Displayed WaterGrid AVL Tree metrics such as root node, height, and balance factor.
8. Verified AVL balance property after all pipeline insertions.

Results and Screenshots:

```
MOVIES INSERTED:
Avatar, Inception, Titanic, Joker, Interstellar

AVL INSERTION PROCESS

1) Inserted Avatar
   Root = Avatar

2) Inserted Inception
   Balance Factor of Avatar = -1
   No Rotation Required

3) Inserted Titanic
   Balance Factor of Avatar = -2
   RR Rotation Applied at Avatar

4) Inserted Joker
   Balance Factor of Titanic = 1
   No Rotation Required

5) Inserted Interstellar
   Balance Factor of Titanic = 0
   AVL Property Maintained
```

```
FINAL AVL TREE

      Titanic
     /      \
  Inception  Joker
   /      \
 Avatar  Interstellar

AVL TREE INFORMATION
-----
Root Movie      : Titanic
Tree Height     : 3
Balance Factor Root : 0
AVL Property    : Maintained

SEARCH OPERATION
-----
Searching for Titanic...

Movie Found

Time Complexity:
Insertion : O(log n)
Search    : O(log n)
```

Time Complexity: WaterGrid AVL Insert, Search, and Delete operations are performed in $O(\log n)$ time due to automatic tree balancing, ensuring rapid pipeline data access even as the water distribution network scales.

Compared to an unbalanced BST, the WaterGrid AVL Tree provides significantly better worst-case performance because the pressure-node tree remains balanced after every insertion and deletion, preventing degradation in water-network query times.

GitHub Link: <https://github.com/ritwikxdevv/DSA-CASE-STUDIES>

Student Signature	Faculty Signature